VoiceShield

AI Voice Security for the Call

Real-time protection against voice phishing and AI deep-voice fraud — during the call, on the device.



Hyungjin Lukas Kim

Founder & CEO · Metacrowd Corp.

Metacrowd.AI

BACKGROUND

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One phone call can erase a lifetime of savings

$7.7B

lost to fraud by Americans aged 60+ in 2025 alone — up 59% in a single year

US mother gets call from ‘kidnapped daughter’ – but it’s really an AI scam

Jennifer DeStefano tells US Senate about dangers of artificial technology after receiving phone call from scammers sounding exactly like her daughter



Scottsdale, AZ

A mother heard her daughter’s AI-cloned voice begging for help — and a $1M ransom demand. Told under oath to the U.S. Senate (2023).

+1,600% surge in deepfake voice-phishing attacks in the U.S. in Q1 2025 vs. Q4 2024

$40B / yr projected annual U.S. losses to AI-enabled fraud by 2027 (Deloitte)

Sources: FBI IC3 (2025); Keepnet (2025); Deloitte Center for Financial Services

THE PROBLEM

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Your data has already leaked — and it calls you back

The last line of defense is the call itself.

Data protection failed
at the source.

Data is already out there

Names, phone numbers, and identity data have been exposed in breach after breach.

Criminals weaponize it

Leaked profiles are compiled into scripts that reference your bank, your family, your life.

It returns as a phone call

A call that sounds completely legitimate — and now, with AI, sounds like someone you trust.

WHY NOW

Anyone’s voice can now be cloned in seconds

2-3 sec

of audio is all it takes to clone a voice

— enough to impersonate your bank, your boss, or your own child.

— After-the-fact response

Blocklists, fraud reports, chargebacks — the money is already gone.

— In-the-moment protection

Detect the scam while the call is happening — before money or data moves.



SOLUTION

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Real-Time AI Voice Scam Detection Service, **'VoiceShield'**

Simultaneously detects in real time whether a voice is AI-generated through technical analysis of voice pattern data and assesses the risk of the call content for criminal intent, assigning a risk score for each category.

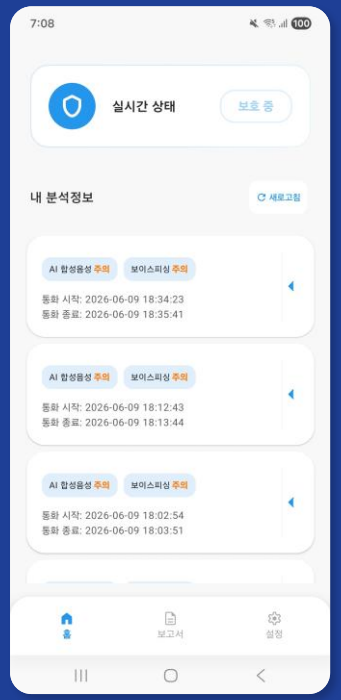
B2C Service (Korean version)



Install from Google Play

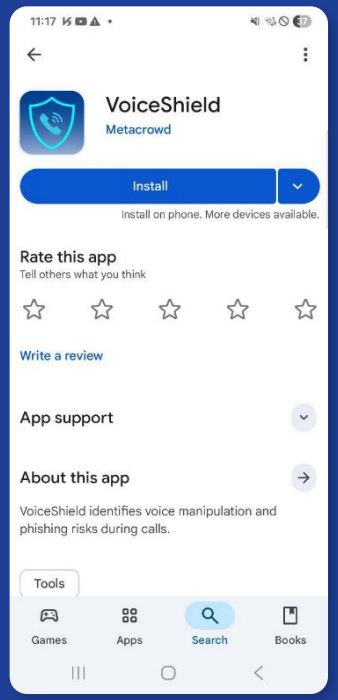


Live risk flags during the call



Per-call analysis history

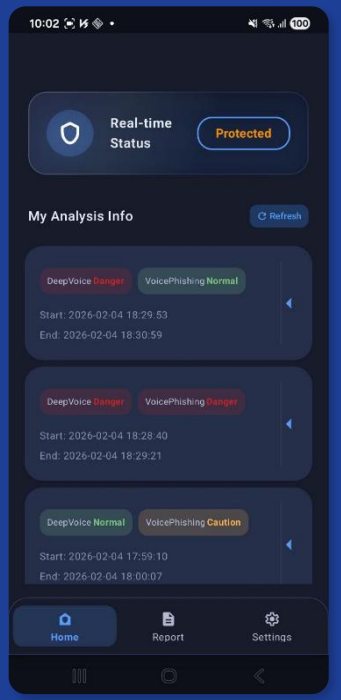
B2C Service (English version)



Install from Google Play



Live risk flags during the call



Per-call analysis history

HOW IT WORKS

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On-device AI that warns before money moves

DVD

Deep-voice detection
via spectrogram phase & time analysis

VAD

Voice activity detection strips silence and noise

STT

Speech-to-text
with speaker separation

SLM

On-device language model reads intent and context

Dual AI engine

CNN-TCN acoustic detection + Tiny-LLM scam analysis, ONNX-optimized for real-time CPU inference on mid-range phones

5–10 seconds

to a clear warning
— before any money is sent or information shared

100% on-device

private audio never leaves the phone;
low battery and storage footprint

KEY DIFFERENTIATOR

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Two signals, one engine — and it stays private

Privacy by design (Full On-device)

Personal data never leaves the device
— servers see only lightweight logs.

Context catches the scam

Voice phishing is caught
through the meaning of the conversation itself.

Acoustics catch the clone

Deep voice is caught through micro-patterns of synthetic audio
— trained on real call data.

Benchmark: DST-NET vs. Reality Defender

Identical conditions · 6 benchmark datasets · 50 samples each

	DST-NET (Ours)	Reality Defender
ROC-AUC ↑	**0.981**	0.942
EER ↓	**0.049**	0.115
FPR@TPR95 ↓	**0.093**	0.143

96% certified detection accuracy	**<1%** false positives	**#1** across all ASVspoof5 categories

Source: ASVspoof5 Challenge; Y. Zhu et al., 2024 (arXiv); KOTCA certification (2025)

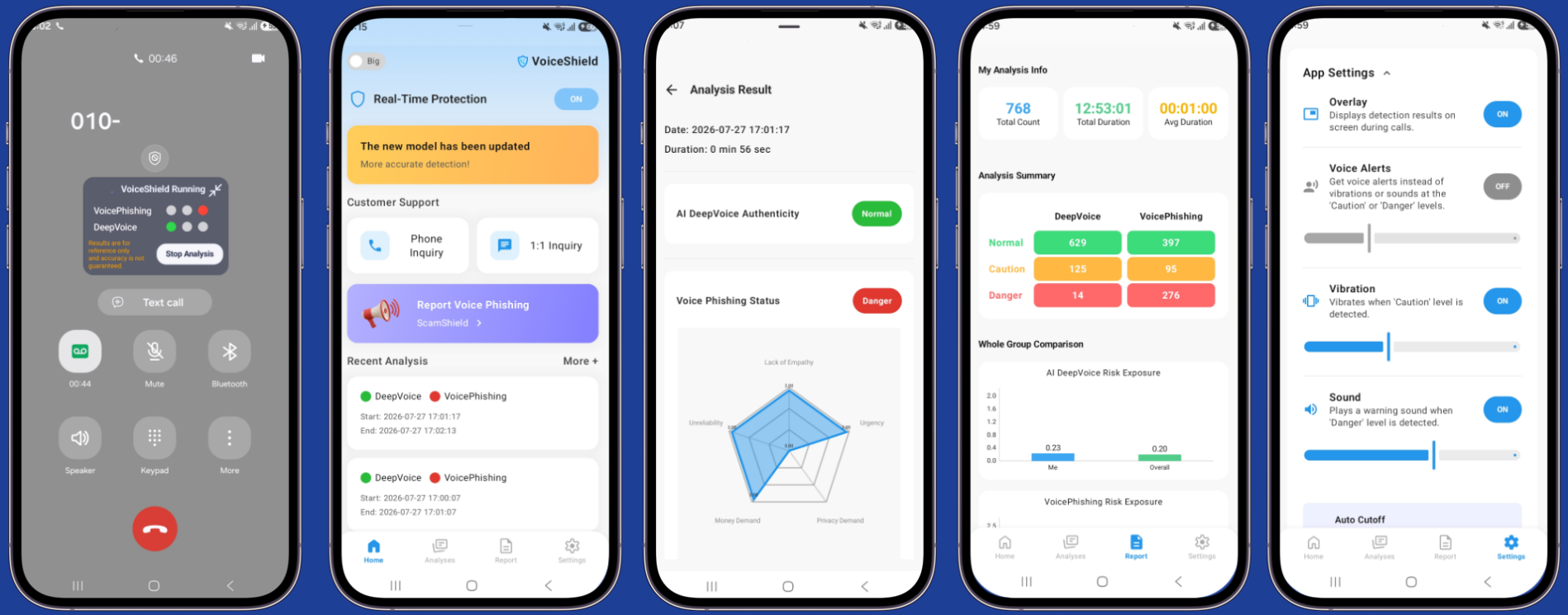
PRODUCT

Metacrowd.AI

A clear warning, in the moment that matters

When a suspicious call comes in

VoiceShield flags voice-phishing and deep-voice risk right on the screen — yellow for caution, red for danger.



GLOBAL READY

Metacrowd.AI

Built for the world from day one

Modular multilingual learning

One core detection model learns multiple languages at once.
New markets are added as lightweight language modules
on top of the accumulated model
— no retraining from scratch, no accuracy reset.

Validated in Singapore

— a four-language market — before U.S. entry.

Live

Korean | English | Arabic

Module-ready

Spanish | French | Japanese

TRACTION

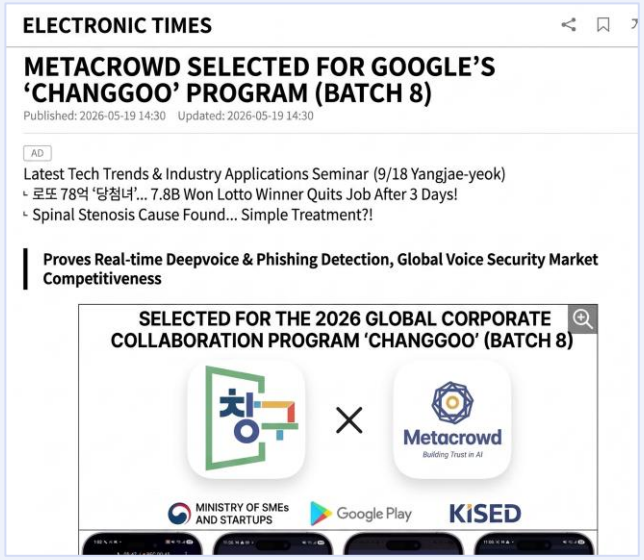
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Already proving it in market

Launched in Singapore

VoiceShield live on Google Play in Singapore (2026) and South Korea (2025)
— accuracy validated in a four-language market

Selected for Google's startup program (2026)


ELECTRONIC TIMES

METACROWD SELECTED FOR GOOGLE'S 'CHANGGOO' PROGRAM (BATCH 8)

Published: 2026-05-19 14:30 Updated: 2026-05-19 14:30

AD

Latest Tech Trends & Industry Applications Seminar (9/18 Yangjae-yeok)
로또 78억 '당첨녀'... 7.8B Won Lotto Winner Quits Job After 3 Days!
Spinal Stenosis Cause Found... Simple Treatment?!

Proves Real-time Deepvoice & Phishing Detection, Global Voice Security Market Competitiveness

SELECTED FOR THE 2026 GLOBAL CORPORATE COLLABORATION PROGRAM 'CHANGGOO' (BATCH 8)

창구 × Metacrowd

MINISTRY OF SMEs AND STARTUPS · Google Play · KISED

Partnership & PoCs

with South Korea's leading internet bank, MVNO operators, and cybersecurity companies.

KakaoBank·Hanabank·CA-GIP
Digital Banking Security

kakaobank · CRÉDIT AGRICOLE · KEB Hana Bank

SKT·Tplus Mobile·KMVNO Association

SK telecom · tplus · KMVNO 한국알뜰통신사업자협회

AhnLab – V3 Mobile Plus & Security

AhnLab

Field-validated impact

Municipal pilot with 147 seniors:
early scam recognition up 23%p,
suspicious-call termination up 24%p

SEOUL · 광주광역시 GWANGJU CITY · 송파구 · 서대문구 SEODAEMUN-GU

Defensible IP

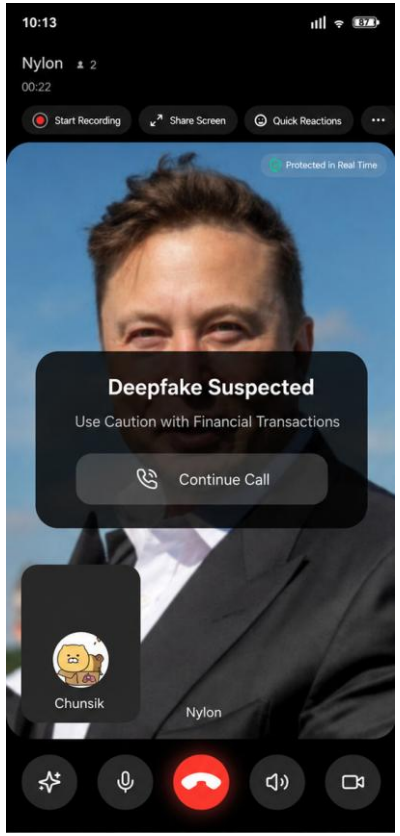
7+ patents on deep-voice & deepfake detection — PCT filed, entering U.S. and Singapore. VOICESHIELD trademark secured in South Korea & Singapore.

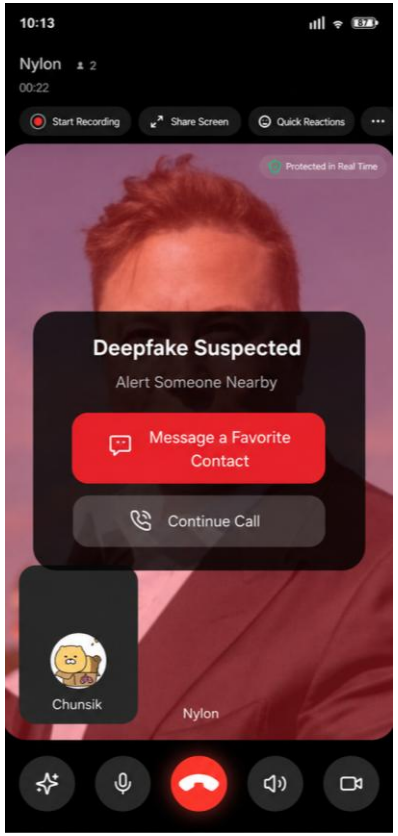
✓ Expansion

Deepfake Detection for Video Calls

When suspicious deepfake content is detected during a video call,

an additional detection engine is activated and user protection measures are initiated





AI Voice Scam Detection + **Video Deepfake Detection**

1. Integrates video deepfake detection with the existing AI voice scam detection engine, expanding protection during video calls.

2. Applies warning and behavioral guidance features to video calls to help prevent user harm.

MARKET OPPORTUNITY

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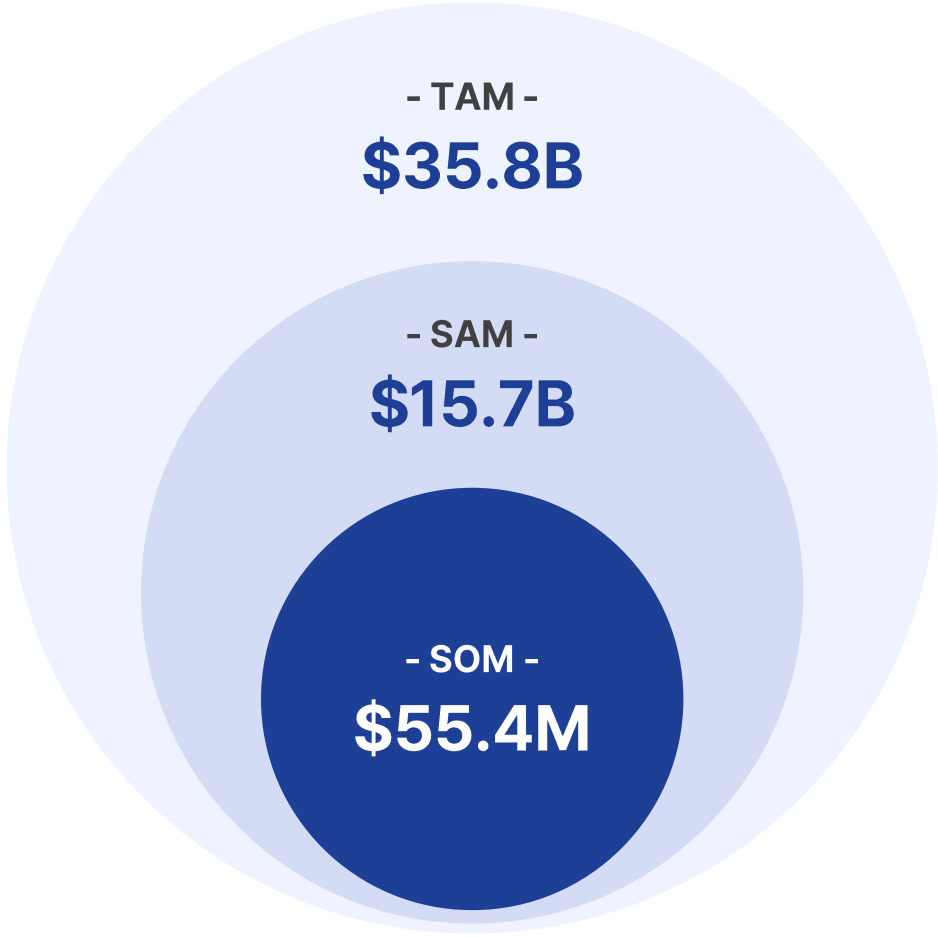
Every smartphone user is a potential customer

U.S. entry (Phase 1, H2 2026)



English-language models already validated; entry via fintech and carrier partnerships.

B2C	Consumer subscriptions — protection for every smartphone user
B2B	APIs, SDKs & licensing for carriers, banks, and security vendors
B2G	Fraud investigation, digital evidence, and defense applications



Deepfake market · Detection market · Initial voice-security target
Source: Navistrat Analytics (2025)

TEAM

Metacrowd.AI

Built to ship core AI fast — and take it global

In-house AI and security expertise, U.S.-native marketing, and legal, data-law, and patent advisors for global compliance.

Hyungjin Lukas Kim

Founder & CEO

Leads U.S. entry, partnerships, and fundraising



- PhD, Management Engineering, KAIST — Information Security, Analytics, Responsible AI
- Associate Professor & Dept. Chair, AI Business Management, Myongji University
- Research Professor & Center Head, Big Data Convergence Research Center, Korea University
- Ex-Data Modeler (KeytalkAI), Data Analyst (Dable)
- BBA & MBA, Korea University · NUS ESP

CORE TEAM

Kyungwook Park

CTO

- 15 yrs in ML & systems — ETRI, LG Electronics, SK M&Service
- PhD candidate (ABD), Computer Science, Hanyang Univ.

Yujin Kim

CMO

- Based in the U.S., leading our expansion on the ground
- ~15 yrs Creative Director & Designer, Manhattan NYC; U.S.-educated

Jinhyun Kim

COO

- Business & strategy planning; IT-service planning at Samsung SDS
- BBA, Korea University

Yerin Park

Business Development

- Voice & sound specialist; 10 yrs in content & education
- BA Seoul National Univ. · MM Manhattan School of Music

Gabin Sung

Front-end Developer

- React Native & Kotlin
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Dongju Kim

AI Modeler

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Jungwook Jeon

Patent · IP

CEO, ZeroOne Patent Law Firm · Seoul National Univ., EE

BUSINESS MODEL & THE ASK

How we earn — and what we're raising

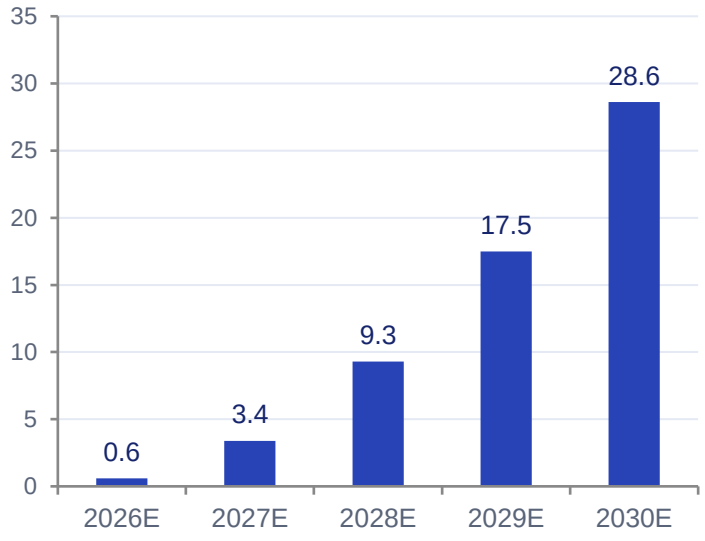
B2C subscriptions

Paid VoiceShield app subscriptions for consumers

B2B / B2G licensing

APIs, SDKs & bundles for carriers, banks, and enterprises

Revenue trajectory (USD millions)



Break-even 2027
51% operating margin by 2030

Converted at ₩1,400 = US$1

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THE ASK

We are entering the U.S. market

and raising our round.

- Capital to scale U.S. go-to-market
- U.S. carrier & bank partnerships
- Senior-protection pilot intros

Voice is the next frontier of security.

Join us in building the most trusted voice shield in the world.

VoiceShield

Building Trust in AI

Metacrowd.AI

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